4 Three plotting compasses are placed in the magnetic field of a bar magnet. The circles in Figure 4.1 show the positions of the plotting compasses.

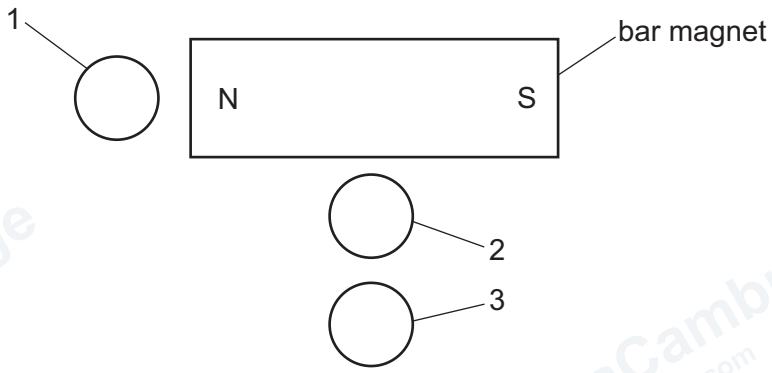

Figure 4.1

(a) State the effect that a magnetic field has on a magnetic pole.

..... [1]

(b) Figure 4.2 shows an arrow that represents the needle in a plotting compass. The needle points in the direction of the magnetic field.

Figure 4.2

On Figure 4.1, draw an arrow in each numbered circle to show the direction in which the compass needles point. [2]

(c) State how the magnetic field at circle 3 is different from the magnetic field at circle 2.

..... [1]

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(d) The bar magnet is moved into a coil of wire as shown in Figure 4.3.

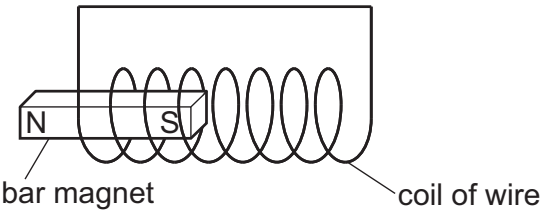

Figure 4.3

Complete the sentences to explain what happens.

As the magnet moves into the coil, the coil experiences a changing magnetic field. An e.m.f. is ..... in the coil. This causes a ..... in the coil which produces a magnetic field around the coil. The magnetic field opposes ..... .

[3]

[Total: 7]

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